

LABORATORY PROJECT REPORT

PLC OVER ETHERNET

EXPERIMENT 7

Section 1

Group 2

Semester 1 2025/2026

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Date of submission: 26th November 2025

Abstract

In this project, an Arduino microcontroller and the OpenPLC Editor are used to construct and test a Programmable Logic Controller (PLC) control system over Ethernet/IP. Developing, simulating, and uploading ladder-diagram logic to operate a basic electromechanical device with push buttons and an LED was the goal. Using common PLC components like contacts, coils, and timer blocks, the experiment started with a simple flashing LED program before moving on to create a Start-Stop control circuit. The ladder logic was built and sent to the microcontroller for real-time execution once the required variables were defined and mapped to the relevant Arduino digital pins. After that, the hardware circuit was built and tested to ensure proper operation. The outcomes validated the usage of PLC-based control using open-source software and microcontroller hardware by showing effective connection between OpenPLC and the Arduino board.

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1.0 Introduction

Because of their dependability, adaptability, and simplicity of programming, Programmable Logic Controllers (PLCs) are widely used in industrial automation to operate equipment, manufacturing processes, and electromechanical systems. PLC development may now be connected with inexpensive microcontrollers like Arduino thanks to the development of open-source tools, enabling students to investigate practical automation principles in a lab setting. In this experiment, ladder logic programs are created, system behaviour is simulated, and an Arduino board serving as the PLC runtime hardware is interfaced with using the OpenPLC Editor as the main software platform.

Starting with a basic LED blinking program and progressing to a Start-Stop control system, the practical tasks entail developing, modelling, and implementing ladder-diagram control circuits. In order to complete these activities, students must construct variables, assign physical addresses based on Arduino pin configurations, and use USB connection to upload the produced program. To verify the developed logic, a physical circuit made up of push buttons, resistors, and an LED is also put together. Students get practical experience with PLC programming, hardware interface, Ethernet/IP-based control ideas, and the integration of microcontroller and software components in simple automation systems by finishing this exercise.

2.0 Materials and Equipment

2.1 Electronic Component

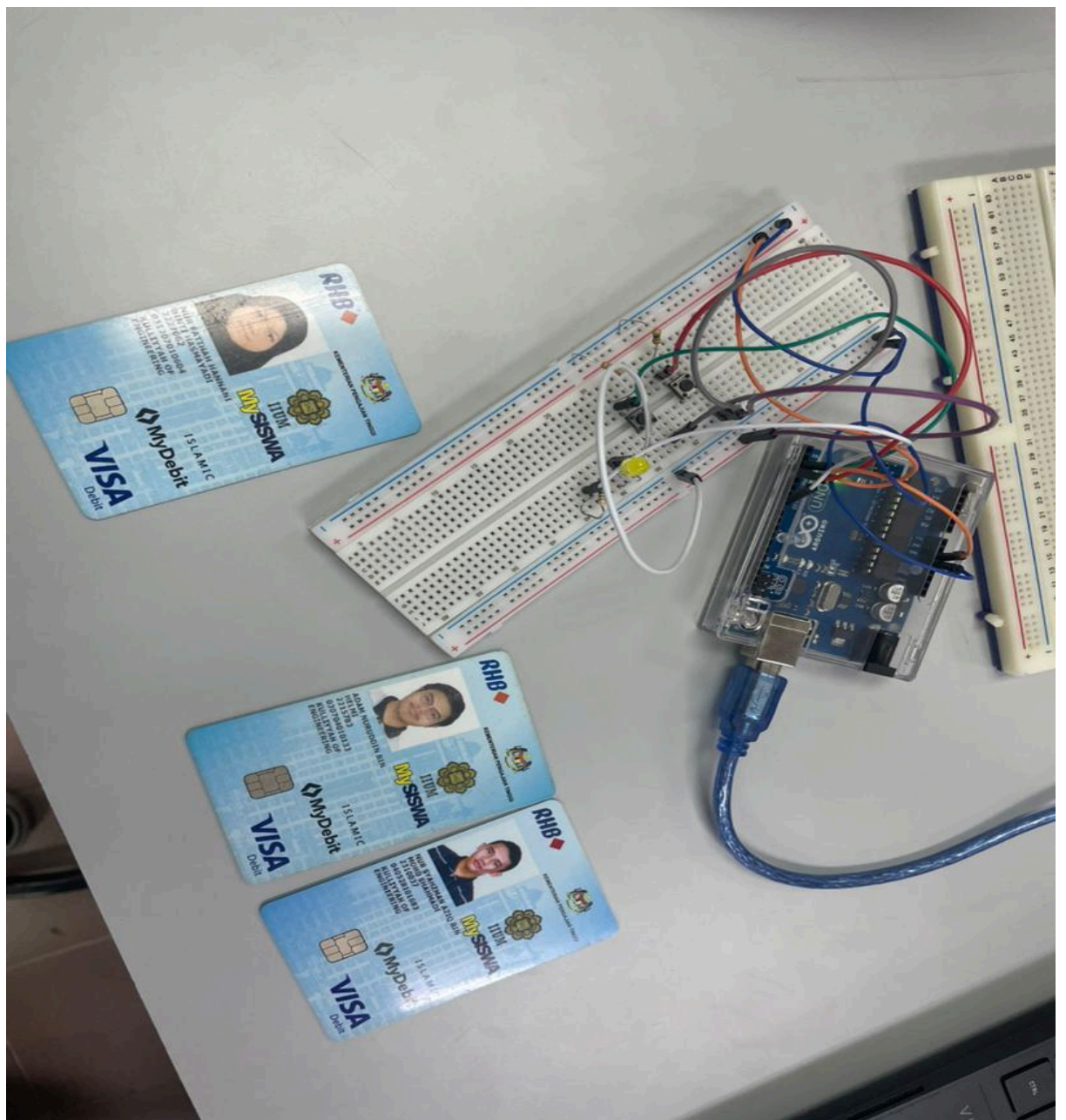
- Arduino Board
- 2 Push Button Switches
- LED
- Resistor
- Jumper Wires

2.2 Equipment and Tools

- OpenPLC Editor Software
- Breadboard

3.0 Experimental Setup

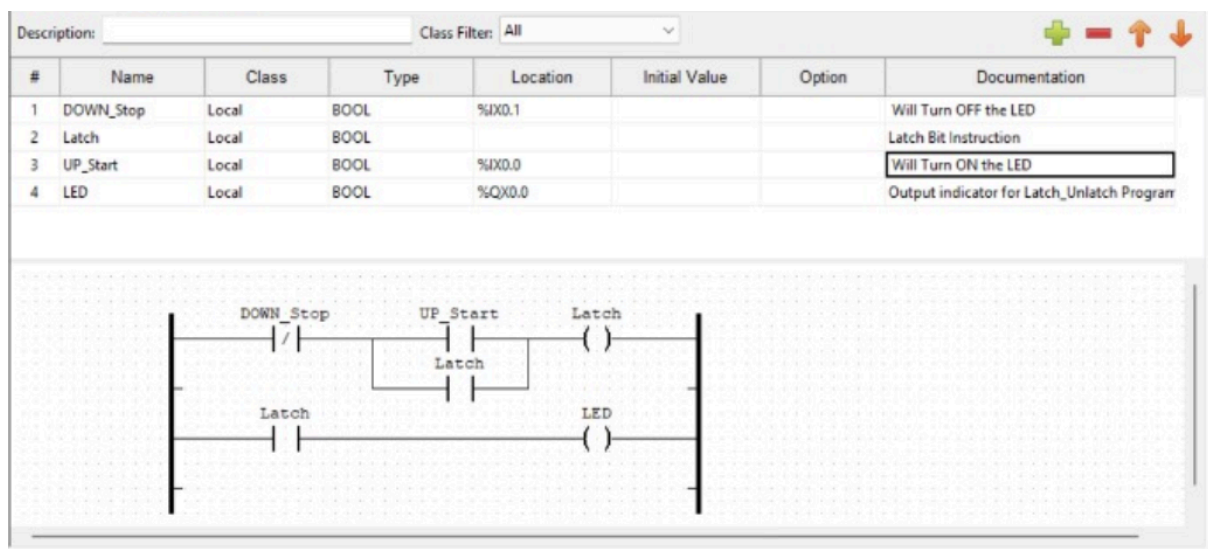
1. Create the ladder diagram.
2. Specify all variables used in the ladder diagram.
3. Compile and simulate the ladder diagram in OpenPLC Editor.
4. Upload the ladder diagram to the Arduino board.
5. Ensure to select correct COM port number and all pin association between the OpenPLC variables and Arduino board.
6. Build the circuit.
7. Test the functionality.



4.0 Methodology

1. Open OpenPLC Editor and create a new ladder diagram project.
2. Declare variables for Start, Stop, and LED output.
3. Draw the Start–Stop ladder diagram using contacts and coil.
4. Compile and simulate the ladder logic to check functionality.
5. Assign each variable to the correct Arduino pin.
6. Connect Arduino to the computer and upload the ladder program.
7. Build the Start–Stop circuit on a breadboard with push buttons and LED.
8. Test the system by pressing Start to turn on the LED and Stop to turn it off.

4.1 PLC constructions



5.0 Data Collection

During this lab, we collected data by observing how our Start–Stop Control Circuit responded when tested using OpenPLC and the Arduino board. We repeatedly pressed the START and STOP buttons and recorded how the LED reacted each time. We checked whether the LED stayed ON after releasing the START button and whether the STOP button successfully turned it OFF. We also verified the I/O pin mapping to make sure the variables in OpenPLC matched the correct Arduino pins. Besides that, we monitored the physical hardware, such as the buttons and wires, to ensure there were no loose connections or unexpected behaviour. Overall, the data we collected focused on the responsiveness, stability, and accuracy of the circuit during operation.

6.0 Data Analysis

Based on the data collected, we analysed the performance of our Start–Stop circuit and found that it behaved exactly like a standard latching system. When we pressed the START button, the LED turned ON and stayed latched due to the holding contact in the ladder logic. Pressing the STOP button immediately broke the latch and turned the LED OFF, showing that the logic executed correctly. This consistency indicates that our ladder program and pin configurations were correctly set up. We also considered potential issues such as input bouncing or unstable output, but we did not observe any major problems during testing. Overall, the analysis shows that both the software simulation and the hardware implementation matched our expected results.

7.0 Results

The results of our experiment show that the Start–Stop Control Circuit worked successfully according to the design. The LED turned ON only when we pressed the START button and remained ON due to the latching contact, while the STOP button reliably turned it OFF. Both the OpenPLC simulation and the real Arduino circuit produced the same outcome, proving that our wiring and programming were correct. The circuit responded consistently even after multiple test cycles, showing that the system was stable and functioning as intended. From these results, we can conclude that the objectives of the lab were fully achieved, and the Start–Stop control concept was implemented effectively.

8.0 Discussion

This experiment successfully demonstrated how the Start-Stop Control Circuit operates using OpenPLC and Arduino simulation. Throughout the testing process, it performed as expected. Each time the START button was pressed, the LED would light up and remain ON due to the touch of the handle programmed into the ladder logic. This confirmed that the basic concept of electrical latching was implemented correctly. The STOP button also consistently interrupted the circuit and turned off the LED, proving that the logic controlling the output behaved reliably under repeated testing.

This experiment also emphasized the importance of connecting the I/O pins correctly. These correct connections were made on the weekly module 'Start-Stop Control Circuit'. This was to ensure that no external factors would interfere with the behavior of the LED. The stable and predictable responses observed throughout the testing indicated that both the programming and wiring were free from major issues of poor connections or time delays.

Overall, the results showed a strong correlation between the simulation environment and the physical implementation. This indicated that OpenPLC provided an accurate representation and indicated that the ladder logic design was robust and well implemented.

9.0 Conclusion

In conclusion, the Start-Stop Control Circuit works as we expected. The LED lights up when we press the START button and stays ON due to the latching logic, and the STOP button successfully turns it off every time. Both the OpenPLC simulation and the actual Arduino setup show the same results, proving that our wiring, pin mapping, and ladder program are correct. This experiment helped us understand how a basic control circuit works, and we achieved all the lab objectives.

10.0 Recommendations

For future improvements, we recommend testing more circuits with components such as relays or motors to see how the Start-Stop logic works in a realistic control system. Also, double-check the wiring connections on the board as there may be things that have been overlooked, such as loose connections and drain connections.

12.0 Acknowledgement

Special thanks to ZULKIFLI BIN ZAINAL ABIDIN & WAHJU SEDIONO for their guidance and support during this experiment.

Student's Declaration

Certificate of Originality and Authenticity

We hereby certify that we are collectively responsible for the work presented in this report. The content reflects our original work, except where specific references or acknowledgements have been made. No part of this report has been completed or submitted by individuals or sources not identified within.

Furthermore, we confirm that this report is the result of a collaborative effort, with contributions made by all group members. The extent of each individual's contribution is documented within this certificate.

We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for marking and this final printed report has been verified by us.

Name: ADAM NURUDDIN BIN HELMI	Read	[/]
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Matric Number: 2215783	Understand	[/]
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Signature: <i>ADAM</i>	Agree	[/]
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Name: NUR FATIHAH HANNANI BINTI HASMAYADI	Read	[/]
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Matric Number: 2227062	Understand	[/]
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Signature: <i>FAT</i>	Agree	[/]
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Signature: <i>AZIQ</i>	Agree	[/]
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